

Question Bank

Unit 1

1. Compare and contrast the working of the human brain with that of a neural network.
2. Describe various types of data handled by deep learning models with examples.
3. Explain the common architectural principles followed in designing deep learning models.
4. What is TensorFlow? Explain its features and significance in deep learning.
5. Describe the role of deep learning in time series, image, and video data analysis.
6. What is Deep Learning? What was the role of deep learning in AlphaGo's success?
7. List any five real-world applications of Deep Learning.
8. Compare and contrast machine learning and deep learning.

Unit 2

1. Describe the structure and working of an artificial neuron with a diagram.
2. Explain different learning rules used in neural networks.
3. What is an activation function? Give examples. Explain its use in neural networks?
4. What is a perceptron? Describe perceptron learning algorithm.
5. Compare linear and non-linear activation functions with suitable examples.
6. Differentiate between single-layer and multi-layer feed-forward networks.
7. Explain the working of a Recurrent Neural Network (RNN) with its applications.
8. Discuss various types of neural network architectures in detail.
9. Describe how neural networks are used in music genre classification.
10. Explain the role of activation functions in deep learning networks with diagrams.
11. Write a short note on how neural networks learn patterns from data.